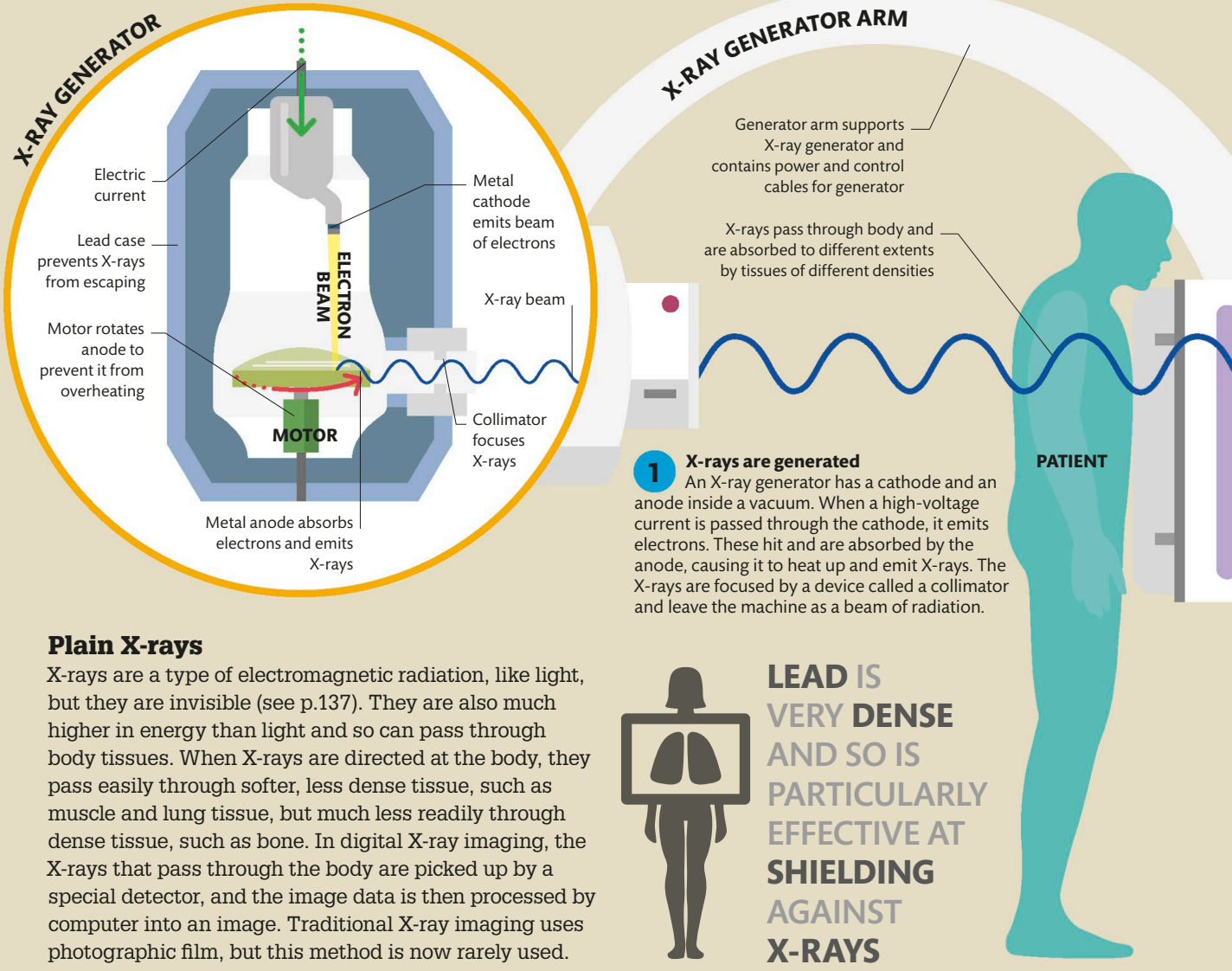
X-ray imaging

The most familiar type of medical imaging, X-rays are used to view internal body tissues and detect disorders, such as fractures or tumors. X-ray imaging is generally simple and painless, although it does involve exposure to radiation.

DO X-RAYS INCREASE THE RISK OF CANCER?

Yes, although the risk depends on the type of X-ray. On average, a single plain X-ray of the chest, limbs, or teeth gives you an additional risk of getting cancer of less than one in a million.

Digital X-ray imaging
The patient is positioned between an X-ray generator and a detector. X-rays from the generator pass through the patient's body to the detector, which converts the X-ray pattern it captures into digital signals. These signals are then processed by computer into an image that is displayed on a monitor.



1 X-rays are generated
An X-ray generator has a cathode and an anode inside a vacuum. When a high-voltage current is passed through the cathode, it emits electrons. These hit and are absorbed by the anode, causing it to heat up and emit X-rays. The X-rays are focused by a device called a collimator and leave the machine as a beam of radiation.

Plain X-rays

X-rays are a type of electromagnetic radiation, like light, but they are invisible (see p.137). They are also much higher in energy than light and so can pass through body tissues. When X-rays are directed at the body, they pass easily through softer, less dense tissue, such as muscle and lung tissue, but much less readily through dense tissue, such as bone. In digital X-ray imaging, the X-rays that pass through the body are picked up by a special detector, and the image data is then processed by computer into an image. Traditional X-ray imaging uses photographic film, but this method is now rarely used.

LEAD IS VERY **DENSE** AND SO IS PARTICULARLY EFFECTIVE AT **SHIELDING** AGAINST **X-RAYS**